

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 20: Cylinder, Cone and Sphere (Surface Area and Volume)

Previous year board questions, topic-wise: 3 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Sphere and hemisphere

1. The volumes of two spheres are in the ratio 27 : 64. Find the ratio of their radii. (a) 3 : 4 (b) 4 : 3 (c) 9 : 16 (d) 16 : 9 **ICSE 2025, 1 mark**

2 Melting and recasting

2. A hollow sphere has an external diameter of 10 cm and an internal diameter of 6 cm. It is melted and made into a solid cone of height 8 cm. Find the radius of the cone.

ICSE 2025, 3 marks

3 Combination of solids

3. A solid wooden capsule is a cylinder with a hemisphere on each end. Its radius is 3.5 cm and the cylindrical part is 14 cm long. The capsule is cut into three separate solids: the cylinder and the two hemispheres. Find the sum of the total surface areas of the three pieces. Take

$$\pi = \frac{22}{7}. \quad \text{ICSE 2025, 4 marks}$$

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